EXP -1

AIM: To simulate and compare the basic resource-allocation methods used in FDMA, TDMA and CDMA, and examine how their assumed user capacity and interference behaviour differ.

Procedure:

1. Run the MATLAB program below.

2. Record the assumed bandwidth, channel width, number of TDMA slots, CDMA processing gain and target SIR.

3. Inspect the printed capacity table and the capacity and CDMA-SIR plots.

4. Inspect the generation comparison table and the conceptual time-frequency resource diagrams.

5. Change at least one parameter for each access method and rerun the program

. 6. Compare the new and baseline results; explain the impact of the changed assumptions.

CODE:

```
clear;

clc;

close all;

% System Parameters

totalBandwidthHz = 2e6;      % Total bandwidth: 2 MHz

fdmaChannelHz = 250e3;       % FDMA channel width: 250 kHz

tdmaSlots = 10;              % Number of TDMA slots

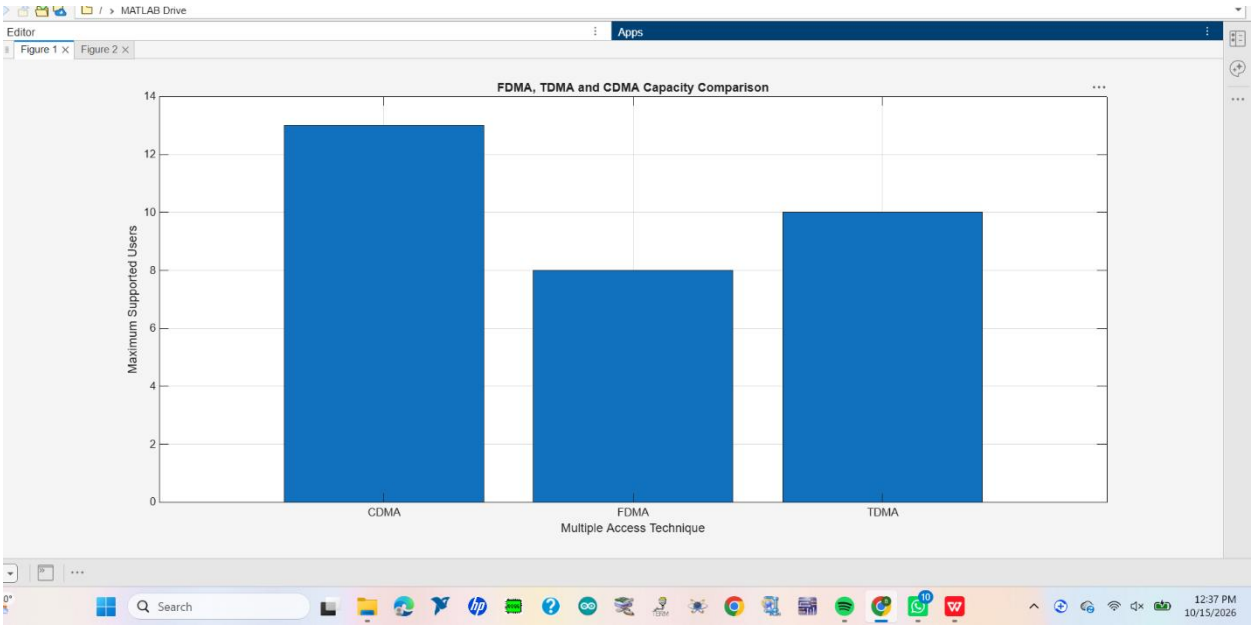
processingGain = 80;         % CDMA processing gain

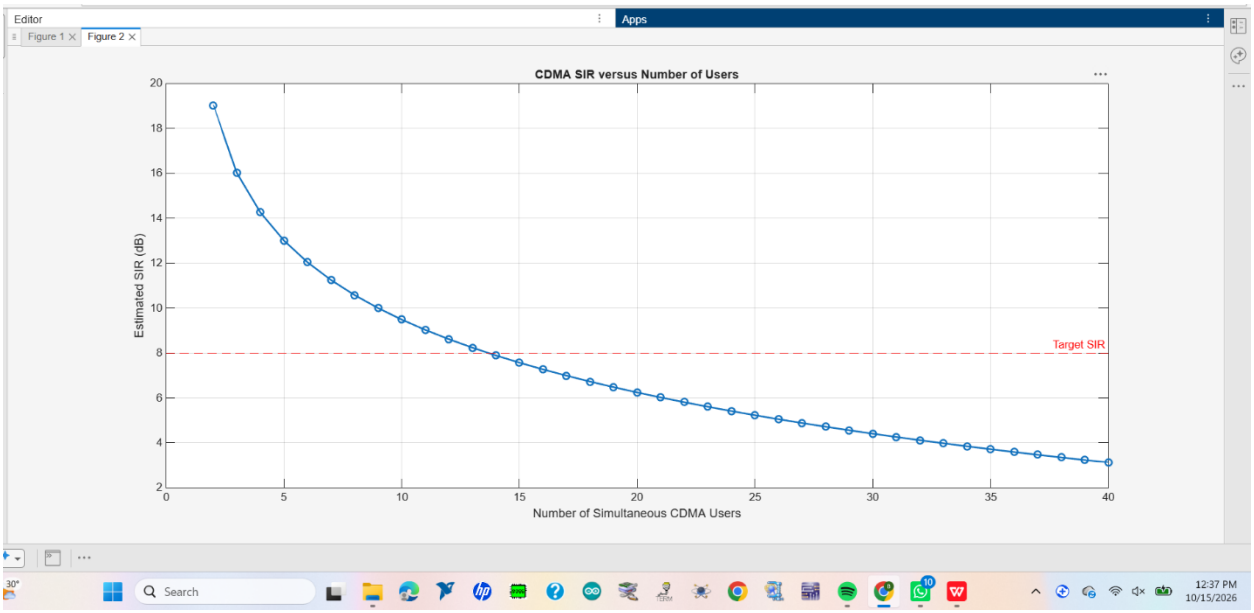
targetSIRdB = 8;             % Minimum acceptable CDMA SIR
```

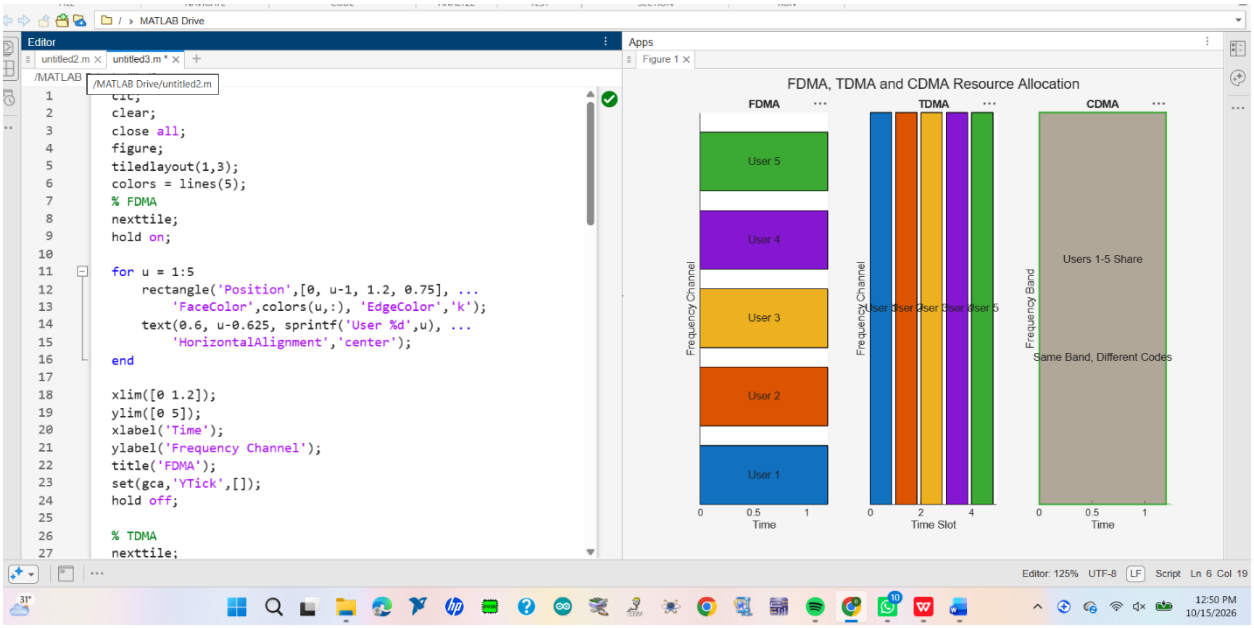
```
% Capacity Estimation
fdmaUsers = floor(totalBandwidthHz / fdmaChannelHz);
tdmaUsers = tdmaSlots;
targetSIR = 10^(targetSIRdB / 10);
cdmaUsers = floor(1 + processingGain / targetSIR);
% Display Results
fprintf('Maximum Supported Users (Estimated)\n');
fprintf('---------------------------------\n');
fprintf('FDMA: %d users\n', fdmaUsers);
fprintf('TDMA: %d users\n', tdmaUsers);
fprintf('CDMA: %d users at a target SIR of %.1f dB\n\n', ...
    cdmaUsers, targetSIRdB);
% Comparison Table
method = ["FDMA"; "TDMA"; "CDMA"];
supportedUsers = [fdmaUsers; tdmaUsers; cdmaUsers];
results = table(method, supportedUsers);
disp(results);
% Plot 1: Capacity Comparison
figure;
bar(categorical(method), supportedUsers);
ylabel('Estimated Maximum Simultaneous Users');
xlabel('Multiple Access Technique');
title('FDMA, TDMA and CDMA Capacity Comparison');
grid on;
% Plot 2: CDMA SIR versus Number of Users
K = 2:30;
```

```
sirLinear = processingGain ./ (K - 1);
sirDb = 10 * log10(sirLinear);
figure;
plot(K, sirDb, 'o-', 'LineWidth', 1.5);
hold on;
yline(targetSIRdB, '--r', 'Target SIR');
xlabel('Number of Simultaneous CDMA Users');
ylabel('Estimated SIR (dB)');
title('CDMA SIR versus Number of Users');
grid on;
hold off;
```

OUTOUT:




Editor
Apps
Figure 1
Figure 2
CDMA SIR versus Number of Users
Estimated SIR (dB)
Number of Simultaneous CDMA Users
Target SIR
20
18
16
14
12
10
8
6
4
2
0
5
10
15
20
25
30
35
40
Search
12:37 PM
10/15/2026


MATLAB Drive
Editor
untitled2.m
untitled3.m *
/MATLAB Drive/untitled2.m
clear;
close all;
figure;
tiledlayout(1,3);
colors = lines(5);
% FDMA
nexttile;
hold on;
for u = 1:5
rectangle('Position',[0, u-1, 1.2, 0.75], ...
'FaceColor',colors(u,:), 'EdgeColor','k');
text(0.6, u-0.625, sprintf('User %d',u), ...
'HorizontalAlignment','center');
end
xlim([0 1.2]);
ylim([0 5]);
xlabel('Time');
ylabel('Frequency Channel');
title('FDMA');
set(gca,'YTick',[]);
hold off;
% TDMA
nexttile;
Apps
Figure 1
FDMA, TDMA and CDMA Resource Allocation
FDMA
TDMA
CDMA
User 5
User 4
User 3
User 2
User 1
Frequency Channel
Time
Time Slot
Frequency Band
Users 1-5 Share
Same Band, Different Codes
Editor: 125%
UTF-8
Script
Ln 6 Col 19
12:50 PM
10/15/2026